

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Faculty of Electrical and Electronics Engineering



Microwave Engineering and Systems

EE5430

Assignment 1

Transmission-Line Parameters and Wave Propagation

Student: Vu Tien Giang

Student ID: 2570188

Report version: Phasor method and engineering checks

September 2026

Contents

1 Preliminaries	2
2 Problem 1	2
2.1 Complex line parameters	2
3 Problem 2	3
3.1 Exact solution	3
3.2 Low-loss model	3
3.3 Velocity and wavelength	4
4 Problem 3	4
4.1 Forward-wave phasors	4
4.2 Electrical length and load phase	5
5 Problem 4	5
5.1 Matched-load fields	5
5.2 Open-circuit case	6
Conclusion	6

1 Preliminaries

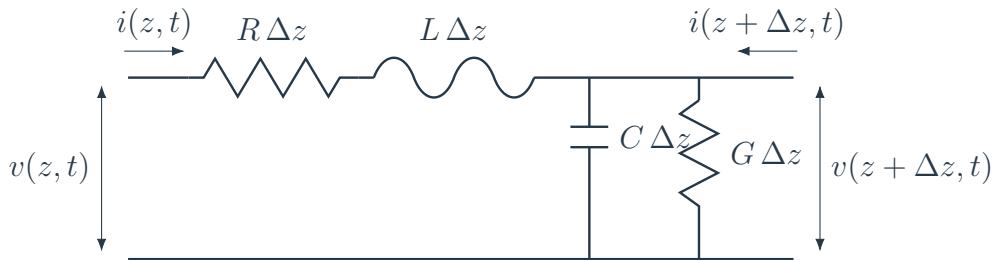
All phasors in this report use $e^{j\omega t}$. Thus a forward wave travelling in the $+z$ direction contains $e^{-\gamma z}$. For a uniform line,

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}, \quad \gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}. \quad (1.1)$$

The principal square roots are selected so that $\alpha > 0$ and $\Re\{Z_0\} > 0$. Peak (not RMS) amplitudes are used in the time-domain expressions.

2 Problem 1

2.1 Complex line parameters



Convert all data to SI units:

$$(R, L, G, C) = (1.5, 10 \times 10^{-9}, 0.4 \times 10^{-3}, 0.2 \times 10^{-12}), qquad f = 2 \times 10^9 \text{ Hz}.$$

At $\omega = 1.256637061 \times 10^{10} \text{ rad/s}$, define

$$Z' = R + j\omega L = 1.5 + j125.663706, \quad Y' = G + j\omega C = 0.0004 + j0.002513274.$$

Then

$$\begin{aligned} \frac{Z'}{Y'} &= 4.884148 \times 10^4 + j7.179858 \times 10^3, \\ Z'Y' &= -0.31522734 + j0.05326548. \end{aligned}$$

Taking the principal square roots,

$$Z_0 = 221.629712 + j16.196087 \, \Omega, \quad (2.1)$$

$$\gamma = 0.047946678 + j0.563494654 \text{ m}^{-1}. \quad (2.2)$$

In polar form, $Z_0 = 222.220706 \angle 4.179588^\circ \, \Omega$. The requested wave quantities follow directly:

$$\alpha = 0.047946678 \text{ Np/m} = 0.41646485 \text{ dB/m}, \quad (2.3)$$

$$\beta = 0.563494654 \text{ rad/m}, \quad (2.4)$$

$$\lambda = \frac{2\pi}{\beta} = 11.1503903 \text{ m}. \quad (2.5)$$

Check: Multiplying the phasor by $e^{-\alpha z}$ shows that the amplitude decreases by $e^{-0.04795} = 0.95318$ over one metre, equivalent to $20 \log_{10}(1/0.95318) = 0.4165 \text{ dB}$.

Answer. $Z_0 = 222.22 \angle 4.18^\circ \Omega$ and $\gamma = (0.04795 + j0.56349) \text{ m}^{-1}$. Hence $\lambda = 11.1504 \text{ m}$.

3 Problem 2

3.1 Exact solution

For the coaxial line,

$$Z' = 4 + j785.398163 \Omega/\text{m}, \quad Y' = 0.001 + j0.314159265 \text{ S/m}.$$

The exact values are

$$Z_0 = \sqrt{Z'/Y'} = 50.00017478 - j0.04774583 \Omega, \quad (3.1)$$

$$\gamma = \sqrt{Z'Y'} = 0.064999970 + j15.707970430 \text{ m}^{-1}. \quad (3.2)$$

Therefore the exact attenuation is 0.064999970 Np/m or 0.56458974 dB/m .

3.2 Low-loss model

The two small parameters are

$$\varepsilon_R = \frac{R}{\omega L} = 0.005093, \quad \varepsilon_G = \frac{G}{\omega C} = 0.003183.$$

Both are much less than one. The first-order low-loss model yields

$$Z_{0,\text{LL}} = \sqrt{L/C} = 50.000 \Omega, \quad (3.3)$$

$$\alpha_{\text{LL}} = \frac{R}{2} \sqrt{\frac{C}{L}} + \frac{G}{2} \sqrt{\frac{L}{C}} = 0.065000 \text{ Np/m}, \quad (3.4)$$

$$\beta_{\text{LL}} = \omega \sqrt{LC} = 15.70796327 \text{ rad/m}. \quad (3.5)$$

The comparison below uses magnitude for the exact characteristic impedance.

	Exact	Approximate	Approximation error
$ Z_0 (\Omega)$	50.00019757	50.00000000	0.000395%
$\alpha (\text{Np/m})$	0.064999970	0.065000000	0.0000456%
$\beta (\text{rad/m})$	15.70797043	15.70796327	0.0000456%

The approximation is more than adequate at 500 MHz.

3.3 Velocity and wavelength

The exact phase values are

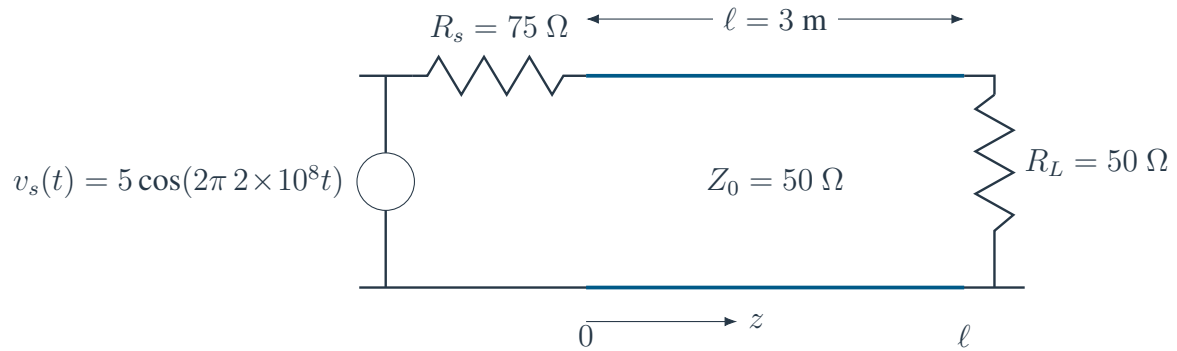
$$v_p = \frac{\omega}{\beta} = 1.99999909 \times 10^8 \text{ m/s}, \quad (3.6)$$

$$\lambda = \frac{2\pi}{\beta} = 0.399999818 \text{ m}. \quad (3.7)$$

The low-loss equations return $2.00000000 \times 10^8 \text{ m/s}$ and 0.400000000 m , respectively.

Answer. $Z_0 \approx 50 \Omega$, $\alpha \approx 0.065 \text{ Np/m}$, $\beta \approx 15.708 \text{ rad/m}$, $v_p \approx 2.00 \times 10^8 \text{ m/s}$, and $\lambda \approx 0.400 \text{ m}$.

4 Problem 3



4.1 Forward-wave phasors

Since $R_L = Z_0$, the reflection coefficient is zero:

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} = 0.$$

The line input also equals 50Ω . Thus

$$\tilde{V}(0) = 5 \frac{50}{75 + 50} = 2 \angle 0^\circ \text{ V},$$

$$\tilde{I}(0) = \frac{\tilde{V}(0)}{50} = 0.04 \angle 0^\circ \text{ A}.$$

The wavelength is $v_p/f = 1 \text{ m}$, giving $\beta = 2\pi \text{ rad/m}$. Therefore

$$\tilde{V}(z) = 2e^{-j2\pi z} \text{ V}, \quad (4.1)$$

$$\tilde{I}(z) = 0.04e^{-j2\pi z} \text{ A}. \quad (4.2)$$

Returning to real time,

$$v(z, t) = 2 \cos(4\pi \times 10^8 t - 2\pi z) \text{ V}, \quad (4.3)$$

$$i(z, t) = 0.04 \cos(4\pi \times 10^8 t - 2\pi z) \text{ A}. \quad (4.4)$$

4.2 Electrical length and load phase

The normalized length and electrical length are

$$\frac{\ell}{\lambda} = 3, \quad \theta = \beta\ell = 2\pi(3) = 6\pi \text{ rad} = 1080^\circ.$$

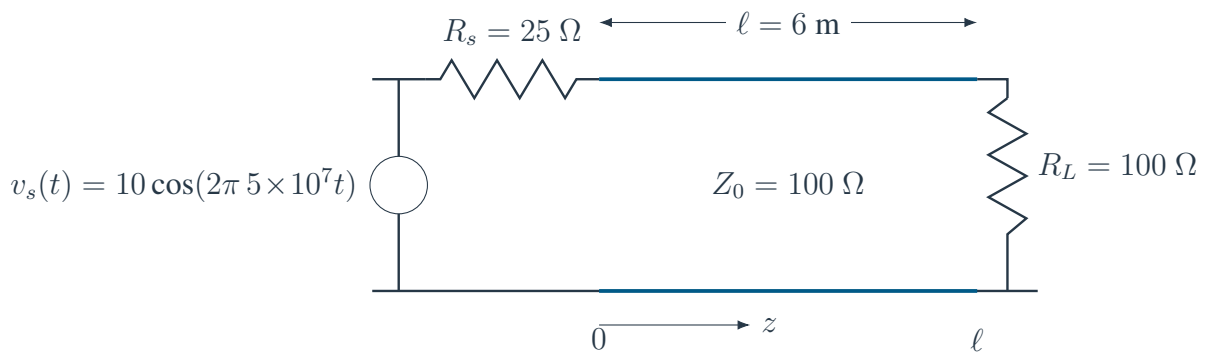
Modulo 360° , the load phase is 0° . Consequently,

$$v_L(t) = 2 \cos(4\pi \times 10^8 t) \text{ V},$$

which is in phase with the ideal source waveform but smaller in amplitude because of the 75Ω – 50Ω voltage division.

Answer. The line contains exactly three wavelengths. The load voltage is 2 V peak and has zero net phase shift relative to $v_s(t)$.

5 Problem 4



5.1 Matched-load fields

The air-filled line has

$$\lambda = \frac{3 \times 10^8}{5 \times 10^7} = 6 \text{ m}, \quad \beta = \frac{2\pi}{6} = \frac{\pi}{3} \text{ rad/m}, \quad \ell = \lambda.$$

With $R_L = Z_0$, the input is 100Ω and

$$V_0^+ = 10 \frac{100}{25 + 100} = 8 \text{ V}, \quad I_0^+ = 0.08 \text{ A}.$$

It follows that

$$v(z, t) = 8 \cos\left(\pi \times 10^8 t - \frac{\pi}{3} z\right) \text{ V}, \quad (5.1)$$

$$i(z, t) = 0.08 \cos\left(\pi \times 10^8 t - \frac{\pi}{3} z\right) \text{ A}. \quad (5.2)$$

The average power delivered to the matched load is

$$P_L = \frac{1}{2} \frac{|V_0^+|^2}{R_L} = \frac{8^2}{2(100)} = 0.320 \text{ W}. \quad (5.3)$$

5.2 Open-circuit case

If the load is open, $\Gamma_L = +1$; a backward wave must be added. Since $\ell = \lambda$, the input repeats the open circuit and the source current is zero. Hence $\tilde{V}(0) = 10$ V. Equal forward and backward wave amplitudes of 5 V satisfy this condition:

$$\tilde{V}(z) = 5e^{-j\beta z} + 5e^{j\beta z} = 10 \cos(\beta z), \quad (5.4)$$

$$\tilde{I}(z) = \frac{5}{100} (e^{-j\beta z} - e^{j\beta z}) = -j0.10 \sin(\beta z). \quad (5.5)$$

Thus

$$v(z, t) = 10 \cos\left(\frac{\pi z}{3}\right) \cos(\pi \times 10^8 t) \text{ V}, \quad (5.6)$$

$$i(z, t) = 0.10 \sin\left(\frac{\pi z}{3}\right) \sin(\pi \times 10^8 t) \text{ A}. \quad (5.7)$$

The current vanishes at $z = 0$ and $z = 6$ m, as expected for this one-wavelength open-ended line.

Answer. The matched formulas give $P_L = 0.320$ W. They do not remain valid after opening the load: the reflected wave produces a standing-wave voltage $10 \cos(\pi z/3) \cos(\pi 10^8 t)$ V.

Conclusion

The exact lossy-line formulas were used in Problems 1 and 2, while the reflection-free travelling-wave model was used in Problems 3 and 4. The low-loss coaxial approximation agrees with the exact calculation to much better than 0.001%. The open-circuit variant in Problem 4 demonstrates why a load change must be accompanied by a reflection analysis.